

Lab 01 - Digital Information

Make a copy of this document for your work (File-> Make a Copy). Type your answers into the boxes below each question; they will expand as you type. When you are done, download your document as a PDF (File -> Download -> PDF) and submit it to the Lab 01 Assignment in D2L. Solutions will be posted the day after the lab is due and only for students who submit work. Check your answers - learning to self-assess is super important and will help you become a better computer scientist!

1. Eureka! We've invented an entirely new kind of computer! This one is built on trinary logic instead of binary so each trinary-bit, or trit, can actually count the numbers 0,1,2 instead of just 0 and 1. We can now write numbers as a sequence of trits.

For example we can convert the number 37 to 1101 in trits ($1 \cdot 27 + 1 \cdot 9 + 0 \cdot 3 + 1 \cdot 1$)
or the number 24 to 220 in trits ($2 \cdot 9 + 2 \cdot 3 + 0 \cdot 1$)

Now answer the following questions to help a newcomer to trit-computing get acquainted with how they work.

- a. What is the trinary number 21201 in decimal?

- b. What is the decimal number 380 in trinary?

- c. How many individual things can you count with a 5-trit number?

- d. What's the largest number you can represent with 5-trits? (Hint: it's the number 22222)

2. The original ASCII character set required only 7 bits to represent a North American keyboard's characters, numbers, and symbols. Assuming all 128 patterns of 7 bits are already defined:

- a. How many bits (in total) would be needed to represent those patterns plus approximately 75 Western European characters?

- b. What is the maximum number of unique characters that could be represented in that many bits?

****You must answer both questions correctly to receive full credit.**

3. Suppose we wanted to represent the original 128 ASCII characters plus 128 extended characters, plus approximately 17000 characters in Arabic, Armenian, Bengali, Bopomofo, Cyrillic, Devanagari, Georgian, Greek and Coptic, Gujarati, Gurmukhi, Hangul, Hebrew, Hiragana, Kannada, Katakana, Lao, Latin, Malayalam, Oriya, Tamil, Telugu, Thai, and Tibetan, plus approximately 52,000 CJK (Chinese, Japanese, Korean) ideographs.

- a. What is the **minimum** number of bits needed to represent these characters?

- b. What is the **maximum** number of unique characters that could be defined using that number of bits?

****You must answer both questions correctly to receive full credit.**

4. Log into the [Pixelation Widget](#) and make your own image of any size.
- Encode an image of anything you like.
 - Do not simply make an abstract pattern, like a checkerboard. Instead, it should depict something, perhaps your name written out, an icon or logo of some sort.

Paste a copy of your image into the box below:

5. What are the largest dimensions (width and height) of an image we can make with the pixelation widget? Use the [Pixelation Widget](#). What limits the dimensions to this size?